

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

WORK EXPERIENCE

TiMoto AI | Software Engineer

Sep 2025 – Present

- Led backend services, REST APIs, cloud infra, and ML serving for a 3-person engineering team – built and operated **distributed production systems** end-to-end from API layer to cloud deployment
- Designed and maintained **gRPC inter-service API layer** with Django/FastAPI backends; resolved production deadlock under concurrent calls by tracing shared resource acquisition conflicts – **100% evaluation success rate** at **sub-50ms p99**
- Architected **vLLM inference backend with PagedAttention** for LLM serving – eliminated KV cache memory fragmentation under concurrent load; deployed with continuous batching achieving **zero OOM failures** at production traffic
- Designed multi-AZ ECS Fargate with **Terraform IaC**, CloudWatch observability, and circuit breaker auto-rollback – **99.9% uptime**, **44% cost reduction** (\$40–60/month)

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Architected **event-driven TypeScript/React system** with observer pattern decoupling UI state propagation across **25K+ lines** of Chromium – **68% feature delivery acceleration** at **95% test coverage**; shipped to Chrome stable serving **3B+ active users**
- Identified settings navigation as p99 bottleneck at 1,200ms; designed and implemented **lock-free concurrent trie search** eliminating mutex contention – **96% latency reduction**, zero production regressions
- Designed **C++ IPC transport layer** with Protocol Buffers – selected Protobuf over custom serialization for schema evolution and cross-language compatibility; shipped to stable channel at **sub-50ms p99**, **10K+ req/sec**
- Collaborated with Chrome infrastructure team on design documents and code reviews adopted by senior Chrome engineers; changes merged into production branch

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Built stateless **REST API backend on AWS** – selected JWT over session auth for horizontal scalability; auto-scaling policy supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with **PostgreSQL indexing** achieving **sub-100ms** for 10,000+ records

PERSONAL PROJECTS

Pulumi – Contributor | Go, TypeScript, IaC

github.com/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning – under active review by core maintainers of 24K+ star project
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer – studied linearizability guarantees under concurrent operations and partial failures

TECHNICAL SKILLS

Frontend: React, TypeScript, JavaScript, Observer pattern, Event-driven UI, 95%+ test coverage

Backend & APIs: REST APIs, GraphQL, gRPC, Protocol Buffers, Django, FastAPI, Node.js, PostgreSQL, Redis

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Kubernetes, Terraform, Docker, CI/CD, CloudWatch, Prometheus, Grafana

Distributed Systems: Circuit breakers, Exactly-once semantics, Fault tolerance, Kafka, Multi-AZ, Event-driven architecture

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LoRA/QLoRA, PyTorch, LLM-as-a-judge evaluation

Languages: C++, Python, Go, TypeScript, Java, JavaScript, SQL, Bash, Rust

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor